ClipSAT : Math for AI & Data Science

Unit 5.01: Distributions & Maximum Likelihood

Name: _____

Q1. A feature in an AI vision model follows a normal distribution X sim mathcal{N}25, 36 . What is the mean mu and variance sigma^2 of this feature?

[ ] A: mu = 25, ; sigma^2 = 36
[ ] B: mu = 36, ; sigma^2 = 25
[ ] C: mu = 25, ; sigma^2 = 6
[ ] D: mu = 6, ; sigma^2 = 36

Q2. Given n = 4 independent observations from a normal distribution with sum sum_{i=1}^{4} x_i = 12 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

[ ] A: hat{mu}_{text{MLE}} = 3
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 12
[ ] D: hat{mu}_{text{MLE}} = 1.5

Q3. Given n = 8 independent observations from a normal distribution with sum sum_{i=1}^{8} x_i = 24 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

[ ] A: hat{mu}_{text{MLE}} = 3
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 24
[ ] D: hat{mu}_{text{MLE}} = 1.5

Q4. Given n = 12 independent observations from a normal distribution with sum sum_{i=1}^{12} x_i = 36 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

[ ] A: hat{mu}_{text{MLE}} = 3
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 36
[ ] D: hat{mu}_{text{MLE}} = 1.5

Q5. Given n = 16 independent observations from a normal distribution with sum sum_{i=1}^{16} x_i = 48 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

[ ] A: hat{mu}_{text{MLE}} = 3
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 48
[ ] D: hat{mu}_{text{MLE}} = 1.5

Q6. Given n = 20 independent observations from a normal distribution with sum sum_{i=1}^{20} x_i = 60 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

[ ] A: hat{mu}_{text{MLE}} = 3
[ ] B: hat{mu}_{text{MLE}} = 6
[ ] C: hat{mu}_{text{MLE}} = 60
[ ] D: hat{mu}_{text{MLE}} = 1.5

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Q7. Given n = 24 independent observations from a normal distribution with sum sum_{i=1}^{24} x_i = 72 , what is the Maximum Likelihood Estimate MLE hat{mu}_{text{MLE}} ?

[ ] A: hat{mu}_{text{MLE}} = 3

[ ] B: hat{mu}_{text{MLE}} = 6

[ ] C: hat{mu}_{text{MLE}} = 72

[ ] D: hat{mu}_{text{MLE}} = 1.5

Q8. In training a language model token predictor, a binary token occurs k = 3 times across n = 10 independent trials. What is the Maximum Likelihood Estimate hat{p}_{text{MLE}} of the token probability?

[ ] A: hat{p}_{text{MLE}} = 0.300

[ ] B: hat{p}_{text{MLE}} = 0.450

[ ] C: hat{p}_{text{MLE}} = 0.150

[ ] D: hat{p}_{text{MLE}} = 0.700